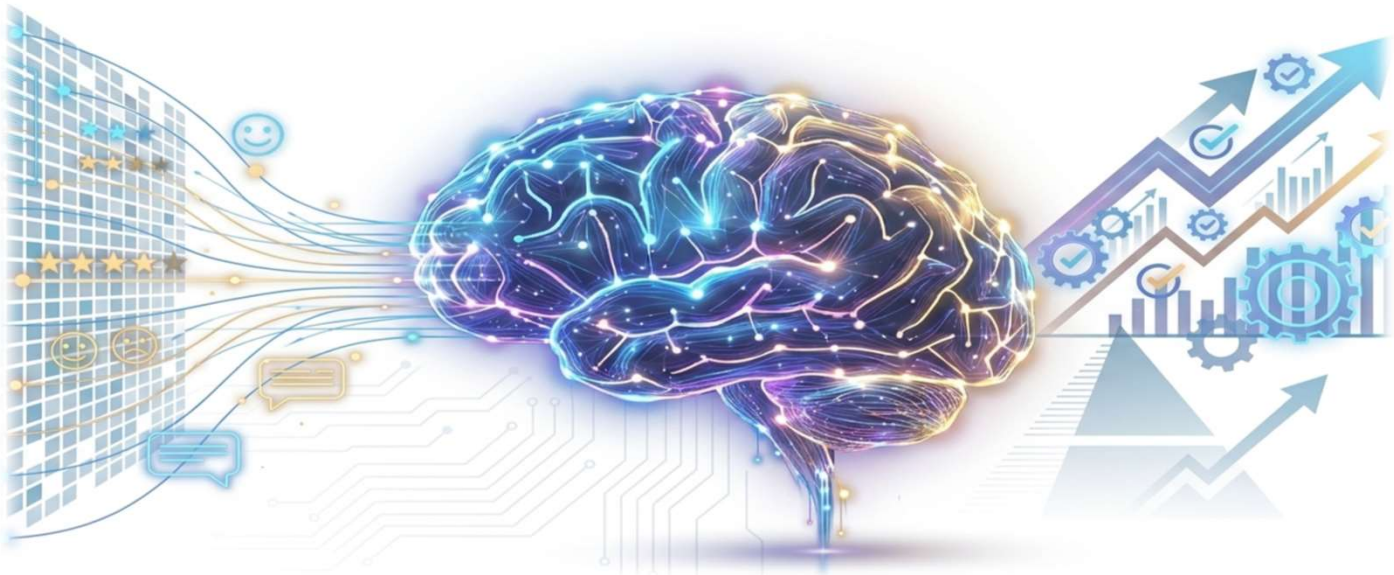
Case Studies in AI for Business

Customer Sentiment Analysis Using AI

Case Study on Fawry Plus



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Fawry Plus: Business Case Overview

Introduction

Fawry Plus was selected as an ideal business case because it represents a clear, practical branch-based retail service, where :

- ✓ Operational Efficiency
- ✓ Customer Experience
- ✓ Real and Measurable Data

Company and Operations Context

Fawry Plus provides assisted financial services at **100-400 customers** per branch daily.

100+ Branches

100-400 Customers Daily

QR Code Surveys

Bill Payments & Financial Services

Customer Reception → Request Verification → Transaction Execution → Confirmation & Feedback

Branch performance

- Post-service QR code surveys
- Identifies service bottlenecks
- Gathers real, measurable survey data impacting operations.

Wait Times

Staff Interaction Issues

System Performance Problems

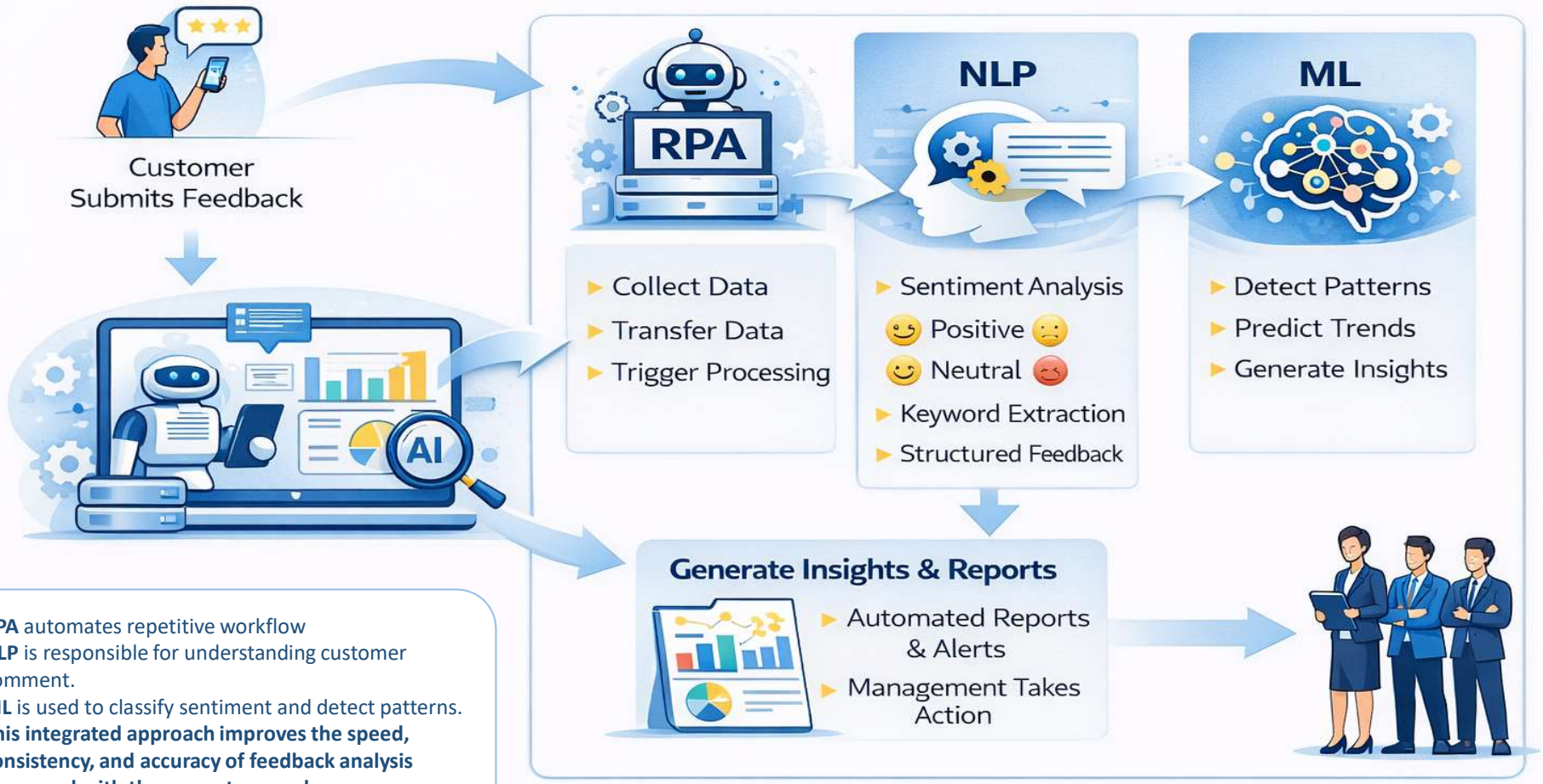
Staff Interaction

Business Problem

The main business problem in this case lies in the ineffective utilization of customer feedback collected through QR-based surveys

At Fawry Plus branches. Although the organization successfully captures a large volume of customer responses, the current process relies heavily on manual analysis using Excel sheets, which significantly limits the ability to extract meaningful insights so it is important to be considered from operational, strategic and financial perspective.

AI Tools for Customer Feedback Analysis



- **RPA** automates repetitive workflow
- **NLP** is responsible for understanding customer comment.
- **ML** is used to classify sentiment and detect patterns.
- **This integrated approach improves the speed, consistency, and accuracy of feedback analysis compared with the current manual process.**

SDLC methodology for an AI-based Customer Feedback Analysis Solution

1. PLANNING

- **IDENTIFY INEFFICIENCY** of manual analysis
- **DEFINE OBJECTIVES** for speed & accuracy
- **DEFINE DECISION-MAKING** goals

2. ANALYSIS

- **STUDY CURRENT WORKFLOW** (QR, Excel)
- **IDENTIFY KEY REQUIREMENTS**
- **ALIGN** with Organizational Goals

3. DESIGN

- **ARCHITECT AI-ML, NLP, RPA** solution
- **DEFINE DATA FLOW**
- **PLAN INTEGRATION** points

4. DEVELOPMENT

- **CREATE ML MODELS** for pattern detection
- **IMPLEMENT NLP** for text analysis
- **DEVELOP RPA WORKFLOWS**

5. TESTING

- **EVALUATE MODEL ACCURACY**
- **TEST RELIABILITY** under conditions
- **IDENTIFY & CORRECT** inconsistences

6. DEPLOYMENT

- **IMPLEMENT** in business environment
- **PROCESS ACTUAL** feedback
- **GENERATE INSIGHTS** for decisions

CONTINUOUS IMPROVEMENT

7. MAINTENANCE & IMPROVEMENT:
Monitor performance, Refine models, Adapt to changing behavior

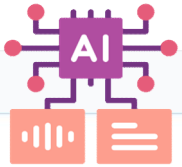
Solution Architecture

1. Input

Customer scans QR code & submits feedback

2. Aggregation

Weekly Excel extract



3. AI Processing

Classifies sentiment, identifies root causes, & predicts risk

4. Action

Instant Dashboard for Managers

Through this Architecture, the organization moves from a slow, manual process to a smart, automated system where insights are generated instantly, decisions are supported by data, and customer experience is continuously improved.

Application of Analytics in the Proposed Solution

In the Fawry Plus case, analytics are applied in a sequence to transform QR survey data into actionable insights to improve customer experience:

Descriptive Analytics

Summarize What Happened

Summarize current feedback to give a clear picture of overall performance.

- **500** Surveys Received
- **350** Satisfied
- **150** Not Satisfied

Diagnostic Analytics

Understand Why It Happened

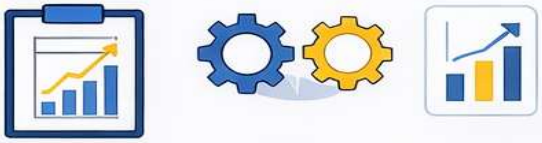
Identify main reasons for dissatisfaction (e.g., waiting time, staff issues, technical problems)

Predictive Analytics

Predict What Could Happen

Forecast future trends in satisfaction and predict potential problems (e.g., increase in waiting time conplaints)

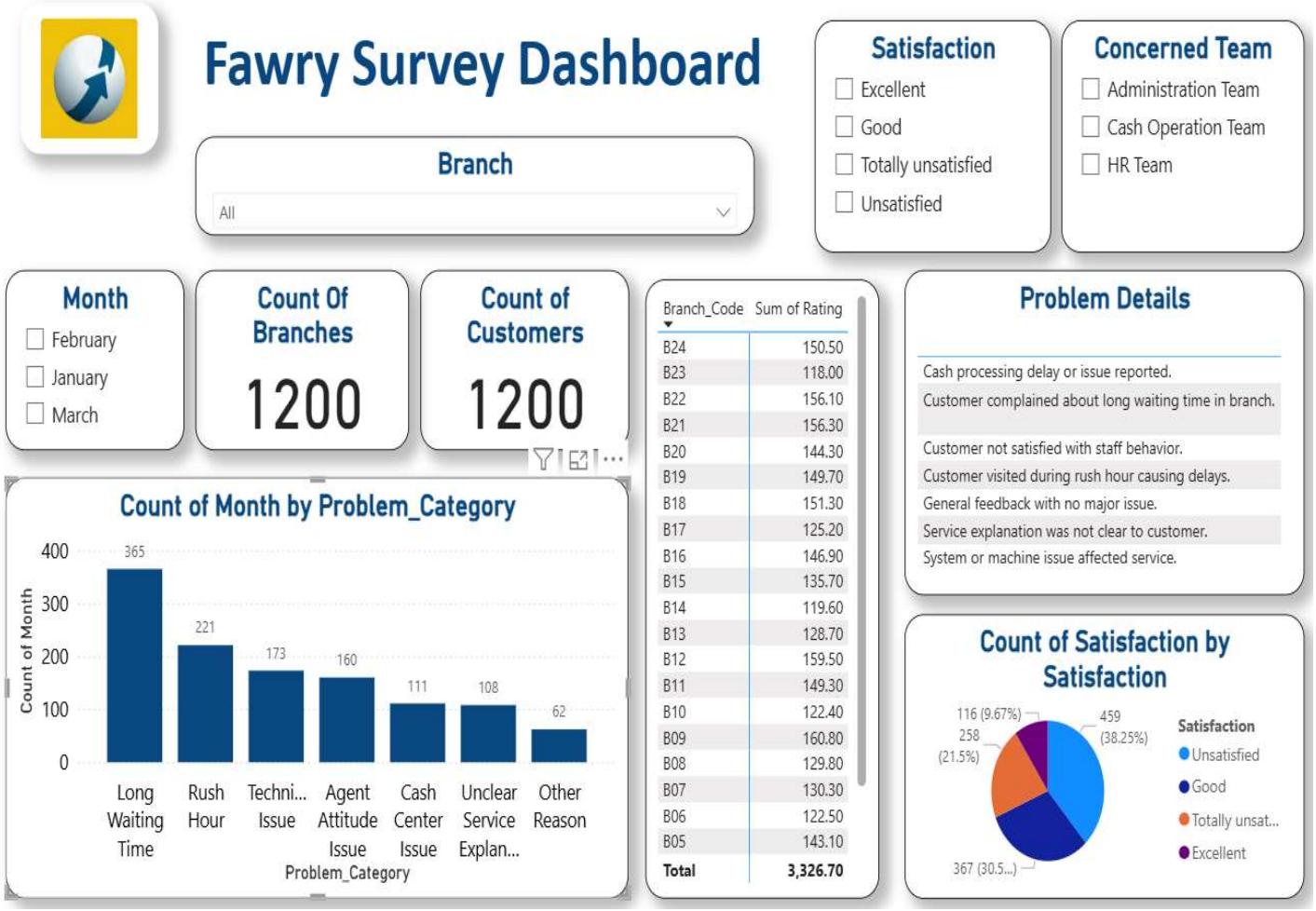
Suggest concrete actions to address identified issues (e.g., add more staff. upgrade systems).



Analytics Application

By combining the mentioned types of analytics changes the QR survey from just a way to collect data into something actually useful for the business support decision making

Helps in understanding customer feedback better by spotting repeated issues, grouping similar comments, and noticing trends over time. This makes it easier for management to take decisions based on real data instead of guessing, which supports continuous improvement, faster response, and a better customer experience at Fawry Plus decision-making



IMPROVEMENT COMPARISON: BEFORE VS. WITH FAWRY CX OPTIMIZER APP

SEQUENTIAL & SLOW — BEFORE AI SOLUTION (MANUAL WORKFLOW)

WORKFLOW & EFFORT

40 STAFF HOURS/MONTH

Manual Cleaning, Categorization, Regional Reporting, Executive Review

LATENCY (TIME TO INSIGHT)

3-4 WEEKS DELAY

Insights generated after issues have affected customers for weeks

ACCOUNTABILITY & IMPACT

AMBIGUOUS

(Misrouted Investigations)

Slowness complaints unclear (System vs. Staff), leading to persistent, recurring issues without targeted intervention.

INCREASED EXPOSURE TO RECURRING ISSUES

WITH FAWRY CX OPTIMIZER (AI-DRIVEN APP) — AUTOMATED & FAST

WORKFLOW & EFFORT

AUTOMATED

(SMALL REVIEW WINDOW)

Single Workflow: Auto-Cleaning, Categorization, Aggregstion of ~1.8K 2K responses/month

LATENCY (TIME TO INSIGHT)

NEAR REAL-TIME

Insights generated shortly after data availability for faster response cycles

ACCOUNTABILITY & IMPACT

CLEAR & ACCURATE

(Data-Correlated Classification)

Improved accuracy in classifying slowness complaints via Data Center report correlation, shortening resolution time.

FASTER RESPONSE, REDUCED RESOLUTION TIME

SURVEY

DATA CENTER REPORT

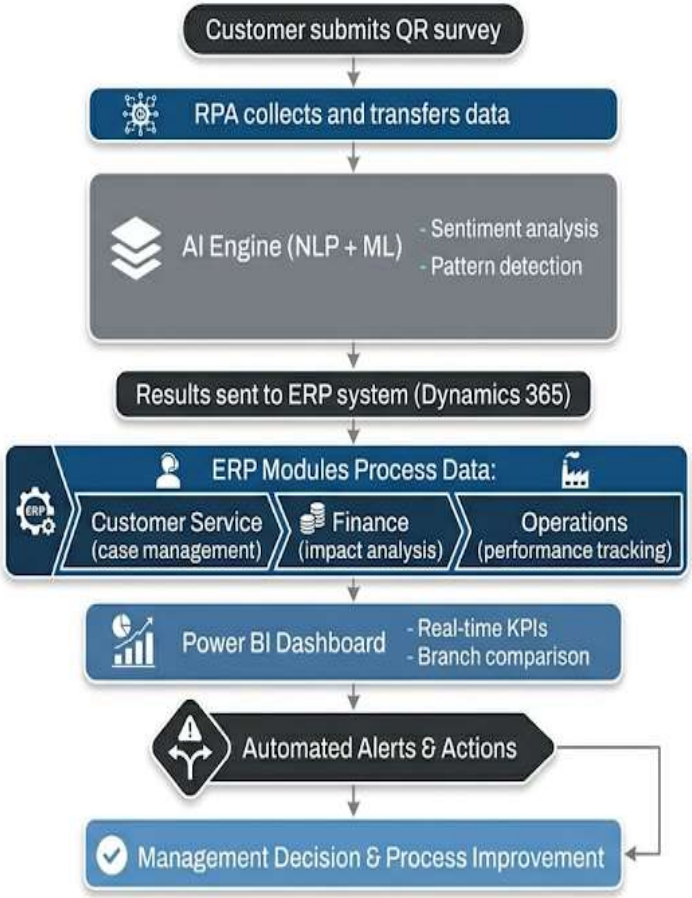
TECHNICAL ISSUE

OPERATIONAL ISSUE

OVERALL IMPROVEMENT MEASURED BY REDUCED ANALYSIS EFFORT, FASTER RESPONSE CYCLES, AND CLEARER ACCOUNTABILITY (NOT SPECULATIVE FINANCIAL ROI).

Integration Between AI and ERP Systems

- Customer feedback is collected through QR surveys
- RPA transfers the data into the analytics environment
- NLP and ML analyze sentiment and classify feedback
- The results are integrated into ERP dashboards (Dynamics 365, Power BI)
- Insights are linked with operational data such as branch performance and service metrics



Risks, Ethical Issues & Limitations

Category	Risk / Issue	Impact	Mitigation
Data Privacy	Misuse of customer data	Legal penalties, loss of trust	Apply GDPR, anonymization, *secure storage*
Ethical Risk	AI bias in sentiment analysis	Unfair or inaccurate decisions	*Use balanced data, continuous model training*
Compliance	Regulatory violations (GDPR, AML)	Fines, operational restrictions	*ERP audit trails, compliance policies*
Technology	System integration complexity	*Delays, higher implementation cost*	*Proper planning, phased implementation*
Adoption	Employee resistance to change	Low system usage	*Training programs, change management*
Data Quality	*Poor or incomplete data*	Low accuracy of results	*Data cleaning, validation processes*
System Limits	*AI misunderstanding language*	*Incorrect classification*	*Model enhancement, human review*
Security	*Unauthorized system access*	Data breaches	*ISO 27001 controls, access control*

Strategic Improvements & Recommendations

Expand Data Sources

Include social media, mobile app data for comprehensive analysis

Enhance Machine Learning Models

Use larger datasets, continuously update algorithms

Improving Real-Time Analytics

Implement live monitoring systems for faster decision-making

Offer Employee Training

Provide training to adapt to new AI and ERP systems

Ensure Data Governance

Enforce strong data security and compliance practices

Thank You